

Making it count: an introduction to the theory of functions

Tom Coleman

Summary

In the third of four linked presentations, this session takes the material covered in the previous two sessions and looks at it from a more general perspective by introducing the theory of functions. Along the way, you'll learn why and how you use functions in everyday life, how every sequence of numbers is a function in disguise, and why pigeons are so important in mathematics.

The function junction

! Try it yourself 1

How many elements are in this set?

$$X = \{a, c, \spadesuit, d, \heartsuit, \clubsuit, , , b, , , e, f, \}$$

i Answer to try it yourself 1

There are 14, but that's not important. Read on...

The important thing was **how** you counted it. If you counted out loud, you probably said 1, 2, 3, 4, ... all the way up to 14, pointing at each element in turn. This is how you are first taught to count things.

In fact, what you are doing is taking an input from X and assigning it an output from the set of all positive whole numbers. This linking of an input to an output is the idea of a mathematical **function**.

i Function, domain, range

Let X, Y be two sets.

A **function** f from the set X to the set Y is an assignment that maps each element of X to **exactly one** output in Y . It is written as $f : X \rightarrow Y$, and the notation $f(x) = y$

means ' f maps $x \in X$ to $y \in Y$.

Here, the set X is called the **domain** of f and the set Y is called the **codomain** of f .

Two functions f and g are equal if and only if they have the same domain X , codomain Y , and $f(x) = g(x)$ for all $x \in X$.

The important thing is that for some $x \in X$, there is exactly one $y \in Y$ such that $f(x) = y$ - so **one** element of X can't map to **many** elements of Y . The converse is not true; **many** elements of X are allowed to **one** element of Y .

Sequences

In mathematics, the reason why functions are incredibly powerful is because they allow for **comparison** between sets. Any time you want to compare two sets - a function is almost always used. Any time you want to look at growth, or rates of change, or to decide behaviour in the future: you will use a function.

One of the ways that functions are studied in mathematics is the idea of a **sequence**. You saw all about sequences in [Exploration: The hidden sequences of Pascal's triangle](#); how they can be either finite or infinite, how they could have formulas or not.

In fact, every sequence is a function in disguise. To help illustrate this, here is some new mathematical notation; this is a common thing to do in mathematics to help explain certain concepts.

i Special notation for sets

Write $[n]$ to be the set $\{1, 2, 3, \dots, n\}$ of all positive whole numbers from 1 up to n included. So $[4] = \{1, 2, 3, 4\}$ and $[k] = \{1, 2, 3, \dots, k\}$. You can notice that the number of elements of $[n]$ is n

For the (infinite) set of all positive whole numbers, write

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

as the set of **natural numbers**.

For the (infinite) set of all possible numbers - whole, fractions and decimals, positive and negative - write $\mathbb{R}$ to be the set of these numbers (called **real numbers**).

This makes it a lot easier to phrase the following important consideration:

i Sequences and functions

- (a) Any finite sequence of length n of symbols from the set $[k]$ can be viewed as a function $f : [n] \rightarrow [k]$.
- (b) In the reverse direction, any function $f : [n] \rightarrow [k]$ is a finite sequence of length n of symbols from the set $[k]$.
- (c) Any finite sequence of numbers of length n can be viewed as a function $f : [n] \rightarrow \mathbb{R}$.
- (d) Any infinite sequence of numbers can be viewed as a function $f : \mathbb{N} \rightarrow \mathbb{R}$, and any function $f : \mathbb{N} \rightarrow \mathbb{R}$ can be written as an infinite sequence.

! Try it yourself 2

Why are (a) and (b) true? As an initial pointer, think about where the set $[n]$ will appear in your consideration.

i Answer to try it yourself 2

For (a), suppose that you have a sequence of length n of elements from $[k]$; you could say that this is

$$a_1, a_2, \dots, a_n.$$

You can see that for every number i between 1 and n , there is an element a_i of $[k]$ at position i in the list. You can then define a function $f : [n] \rightarrow [k]$ by the assignment

$$f(i) = a_i$$

which sends $i \in [n]$ to a_i . This then determines the sequence $a_1, a_2, \dots, a_n$.

For (b), suppose you have a function $f : [n] \rightarrow [k]$. You can now view this as a sequence of length n in $[k]$ by writing out all of the outputs in order:

$$f(1), f(2), \dots, f(n)$$

which defines a finite sequence of length n with elements in $[k]$.

Parts (c) and (d) follow from similar reasoning.

This means that finite/infinite sequences can be studied through the realms of functions. Here's an excellent result that follows from this.

i Counting number of functions

The total number of possible functions $f : [n] \rightarrow [k]$ is k^n .

i Proof of the formula for number of subsets

You are given that $[n] = \{1, 2, 3, \dots, n\}$, and there is a function $f : [n] \rightarrow [k]$. Write f as a finite sequence of length n with elements from k to get

$$f(1), f(2), f(3), \dots, f(n)$$

So, you can ask yourself - what is $f(1)$? Well, you don't know, as you aren't given the information. However, there are k many **choices** for the value of $f(1)$, because it has to be exactly one element of the set $[k] = \{1, 2, 3, \dots, k\}$. Since it can be any of them, this gives k choices.

Next, you can ask - what is $f(2)$? Again, there are k many choices. You can notice here that the choice of the value of $f(2)$ is completely independent of the choice for the value of $f(1)$. Similarly, there are k many choices for the value of $f(3)$ and this choice is independent of the values of $f(1)$ and $f(2)$.

You can carry on in this way for all the elements of $[n]$. With k choices for each element, and all of the choices being independent of each other, this leads to

$$\underbrace{k}_{\text{what is } f(1)?} \cdot \underbrace{k}_{\text{what is } f(2)?} \cdot \underbrace{k}_{\text{what is } f(3)?} \cdot \dots \cdot \underbrace{k}_{\text{what is } f(n)?} = k^n$$

many choices for the sequence $f(1), f(2), f(3), \dots, f(k)$. Therefore, there are k^n possible functions $f : [n] \rightarrow [k]$.

! Try it yourself 3

Does this seem familiar?

i Note

This is an almost a word-for-word recreation of the proof in [Exploration: How to win at cards - combinations and permutations](#) that there were 2^n many subsets of a set $\{1, 2, 3, \dots, n\}$!

So what's happening here? Well, the idea of deciding whether or not each element of $[n] = \{1, 2, 3, \dots, n\}$ is in some subset of $[n]$ can be modelled as a function $f : [n] \rightarrow [2]$, where $f(i) = 1$ says that i is not in the subset, and $f(i) = 2$ says that i is in the subset. Therefore, every function $f : [n] \rightarrow [2]$ corresponds to a subset of $[n]$, and every subset

of $[n]$ corresponds to a function $f : [n] \rightarrow [2]$.

One-to-one and onto

The set

$$f(X) = \{f(x) \text{ where } x \in X\} \subseteq Y$$

is called the **image** of f . It follows from this that the image of f is a subset of the codomain Y of f .

Where and how?

Pigeons and counting

Version history

v1.0: initial version created 04/26 by Holly Goldsmith as part of a University of St Andrews VIP project.

[This work is licensed under CC BY-NC-SA 4.0.](#)